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# Forecasting Hotspots for Predictive Police Patrolling


SUPERVISOR

Prof. Youness Mechqrane

THESIS DIRECTOR

Prof. Amal El Fallah Seghrouchni

PRESENTED BY

**Ismail Elabbassi**


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OUTLINE

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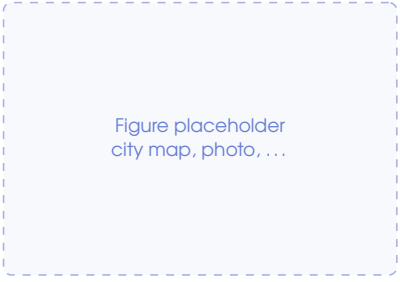

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# 01
# Introduction

Context, motivation and research questions

# Context and motivation

- Placeholder point about the problem setting
- Why forecasting crime hotspots matters
  - Sub point one
  - Sub point two
- Limits of current patrol planning



# Research questions

1. Can we forecast where crime will concentrate in the next period?
2. Which features drive these forecasts?
3. How should patrols be allocated given the forecasts?

**Main contribution**

One sentence that states the contribution of the work.

# 02
# Related Work

Crime hotspot mapping and predictive policing

# Related work

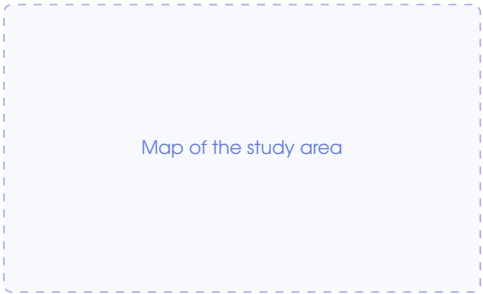
| Approach | Spatial | Temporal | Patrol planning |
|---|---|---|---|
| Kernel density estimation | ✓ | | |
| Self-exciting point processes | ✓ | ✓ | |
| Deep spatio-temporal models | ✓ | ✓ | |
| **This work** | ✓ | ✓ | ✓ |

# 03

# Data & Study Area

Crime records, features and preprocessing

# Dataset




- Source and time span of the records
- Spatial grid resolution
- Features: history, calendar, points of interest
- Train / validation / test split

# 04
# Methodology

Forecasting models and patrol allocation

# Forecasting model

The predicted intensity of cell $i$ at time $t$ is

$$\hat{\lambda}_{i,t} = f_\theta\big(x_{i,t-1}, \ldots, x_{i,t-k}\big),$$

and patrols are allocated to the cells with the highest $\hat{\lambda}_{i,t}$.

**Intuition**

Short explanation of the model in plain words.

# Pipeline

Pipeline diagram: data → features → model → patrol plan

# 05
# Results

Experiments, metrics and comparison

# Main results

**0.82**
Hit rate at 5% area

**+14%**
Over the best baseline

**2.3×**
Predictive accuracy index

## Takeaway

One sentence that the audience should remember from the results.

# 06
# Conclusion

Takeaways, limitations and future work

# Conclusion and future work

## Takeaways

- First takeaway
- Second takeaway

## Future work

- First direction
- Second direction

THE END

# Thank you!

Questions & discussion

**Ismail Elabbassi**
your.email@um6p.ma

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